Tayyab Shahid

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SUMMARY

Fresh Computer Science graduate from University of Education, Lahore, with 1 year of hands-on experience building and deploying Machine Learning and Generative AI projects. Comfortable working across the full ML pipeline — from data preprocessing and model training to building web apps with Streamlit. Developed an AI-powered Sign Language Recognition system as a Final Year Project. Actively seeking an entry-level role in AI/ML.

EXPERIENCE

AI Engineer — TechAbout Private Limited, Lahore JULY 2026 – Present

- Working on building and deploying Machine Learning and Generative AI solutions, including RAG-based pipelines using LangChain, ChromaDB, and Hugging Face Transformers.
- Developing computer vision and deep learning models (OpenCV, MediaPipe, CNN, LSTM) and integrating them into interactive Streamlit web applications for real-time inference.

TECHNICAL SKILLS

Languages & Libraries: Python, NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, NLTK, OpenCV, MediaPipe

Machine Learning: Supervised & Unsupervised Learning, NLP, CNN, LSTM

Generative AI: Hugging Face, LangChain, ChromaDB, RAG, LLaMA, Prompt Engineering

Tools & Deployment: Streamlit, Git, GitHub, Jupyter Notebook, VS Code

PROJECTS

DocChat — AI PDF Question Answer System (RAG) *Python | LangChain | ChromaDB | LLaMA | Streamlit*

- Developed a Retrieval-Augmented Generation (RAG) application that enables users to upload PDF documents and ask questions using natural language.
- Integrated LangChain, ChromaDB, and LLaMA to retrieve relevant document context and generate accurate responses.
- Built an interactive Streamlit interface for seamless document upload and conversational Q&A.

Talking Hands — AI Sign Language Recognition System (FYP) *Python | MediaPipe | CNN | LSTM | Hugging Face | gTTS*

- Designed an AI-powered sign language recognition system capable of translating hand gestures into speech in both English and Urdu.
- Utilized MediaPipe Holistic to capture 387 facial, hand, and body landmarks from live webcam frames for gesture recognition.
- Trained an LSTM-based deep learning model on the WLASL dataset, achieving 72% validation accuracy, with response refinement using Hugging Face Transformers.

Credit Card Fraud Detection *Python | Scikit-learn | SMOTE | Logistic Regression*

- Built a machine learning model to detect fraudulent credit card transactions on a highly imbalanced financial dataset.
- Improved minority class representation using SMOTE before training a Logistic Regression classifier.
- Evaluated performance using Precision-Recall and ROC-AUC, with emphasis on reducing false negatives in fraud detection.

Spam Email Detection System *Python | Naive Bayes | TF-IDF | NLTK | Streamlit*

- Developed an NLP-based spam detection application using text preprocessing and machine learning techniques.
- Implemented text cleaning, tokenization, stemming, and TF-IDF feature extraction before training a Naive Bayes classifier.
- Achieved 97.16% accuracy and 99% precision, and deployed the solution as a Streamlit web application.

House Price Prediction System *Python | Gradient Boosting | Scikit-learn | Streamlit*

- Created a regression model to estimate residential property prices using the King County Housing dataset.
- Trained a Gradient Boosting Regressor that achieved an R^2 score of 0.88 on the evaluation dataset.
- Developed a Streamlit web application that predicts house prices based on user-provided property features.

EDUCATION

Bachelor of Science in Computer Science 2022 – 2026
University of Education, Lahore

Intermediate in Computer Studies 2019 – 2021
Islamia College Railway Road, Lahore

Matriculation in Computer Studies 2016 – 2018
Islamia School Multan Road, Lahore